Uber India Ride Analytics – July 2024

One month of ride-hailing bookings in Bangalore, analysed with SQL, Excel and Power BI. Prathmesh Gondkar, 2025.

Summary

This project takes one month of ride booking data for Bangalore (July 2024) and works through it end to end: cleaning in SQL and Excel, then a Power BI dashboard on top. The questions I wanted to answer were simple ones an operations team would ask: when is demand highest, where is it concentrated, how many requests actually become rides, and which service tiers make the money.

The headline findings: only 62% of requests ended in a completed trip, demand peaks around 8 AM and 8–9 PM, a handful of neighbourhoods account for a disproportionate share of pickups, and the premium Comfort tier earns the most revenue despite similar trip counts across tiers.

A note on the data: the dataset is modelled on the public Ola Bangalore case study, relabelled with Uber service tiers. Locations, volumes and outcomes describe Bangalore. It is a learning dataset, not Uber's actual data.

Tools

- **SQL** for cleaning, type conversion and the aggregations behind the visuals. The queries are in Uber.sql.
- **Excel** for first inspection of the raw file and spot-checking data issues.
- **Power BI** for the model, DAX measures and the dashboard pages.

The data

- **Rows:** 40,534 booking requests, 1–31 July 2024, after removing empty spreadsheet rows.
- **Columns (19):** date, time, booking ID and status, customer ID, vehicle type, pickup and drop locations, driver and customer wait times (V_TAT, C_TAT), cancellation flags and reasons, booking value, payment method, ride distance, driver and customer ratings.
- **Outcomes:** Success, Canceled by Driver, Canceled by Customer, Driver Not Found.
- **Service tiers:** Uber X, Uber XL, Comfort, Uber Black, roughly 10,000 requests each.
- **Cleaning needed:** the text "No" was used as a placeholder in numeric columns for trips that never happened (wait time, fare, payment method), several thousand fully blank rows sat below the data, and location names had spelling variants. Details below.

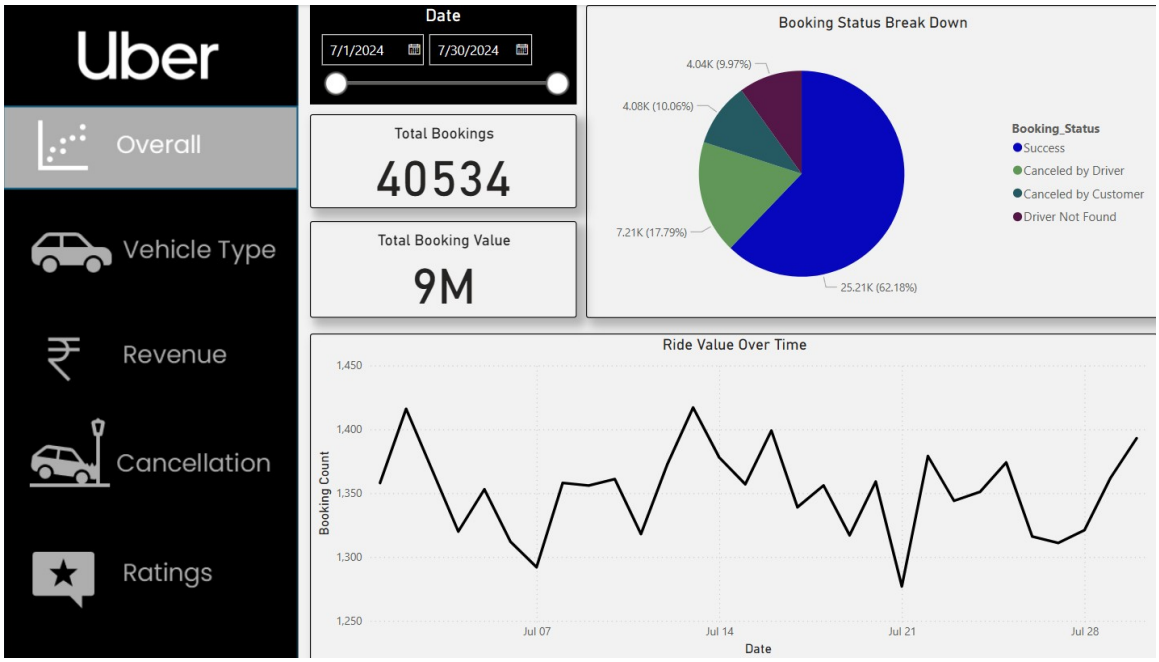
Totals after cleaning: 25,205 completed trips (62%), about ₹9.45 million in fares from completed trips, an average fare near ₹375, and an average distance around 21 km.

Cleaning and SQL

- **Blank rows.** The sheet carried about 62,000 empty rows under the real data. Filtered out on load.
- **"No" placeholders.** Converted to NULL or 0 depending on the column: NULL for payment method, 0 for wait time and fare on cancelled trips. Done with CASE expressions so the rule is visible in the query.
- **Date and time.** Stored as two columns. Combined into one timestamp so hour-of-day and day-of-week could be derived in SQL rather than in the visual layer.
- **Location names.** The source had systematic spelling corruption ("IndiraNogar", "KammaNohalli", "BaNoshankari") that looks like a global find-and-replace gone wrong on the original file. Standardised to the real neighbourhood names so each area aggregates once.

- **Aggregations.** Bookings per hour, per day and per pickup location were produced as SQL summaries and used as the basis for the Power BI visuals, which kept the model light.
- **Checks.** Row counts before and after each step, and totals for completed trips and revenue re-verified after transformation.

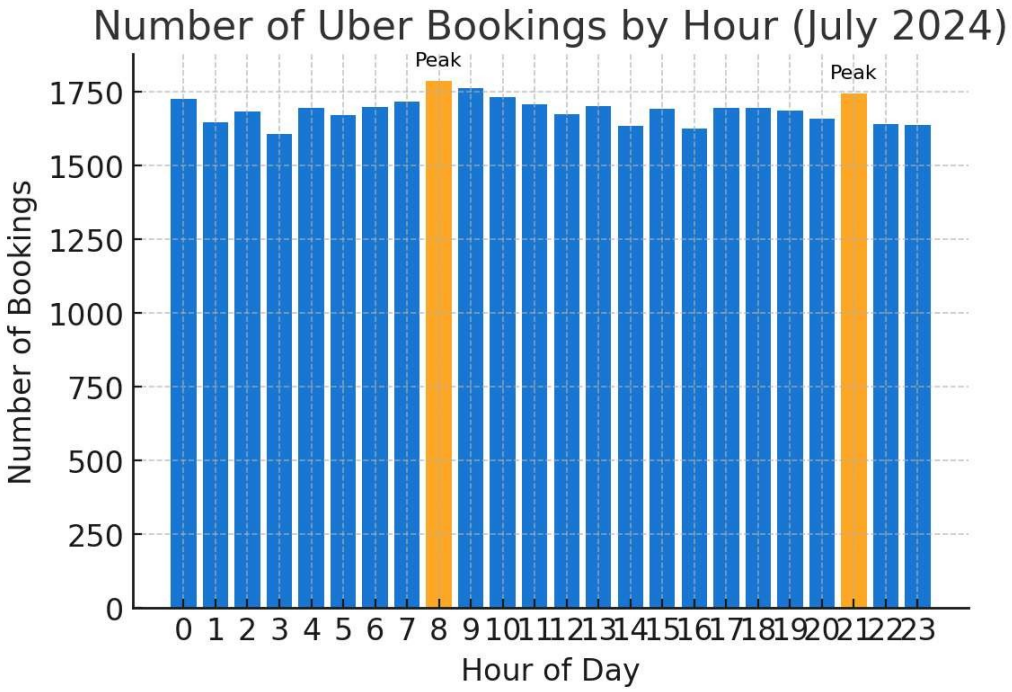
What the data shows



Power BI overview page: total bookings, booking value, outcome breakdown and daily ride value.

Demand is steady, with two peaks

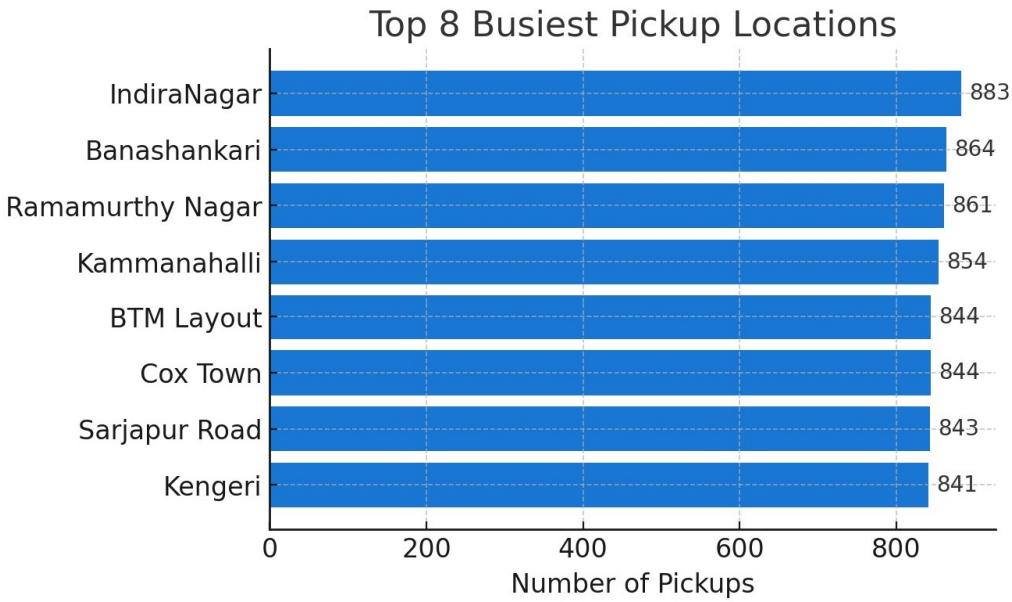
Volume held at roughly 1,300–1,400 requests a day through the month, with no real trend. Weekdays ran slightly ahead of weekends. Hourly, demand climbs from early morning to a peak around 8 AM, dips a little midday, and peaks again at 8–9 PM. Even the quietest hour (around 3 AM) still had over 1,600 requests across the month.



Requests by hour of day. Peaks around 8 AM and 9 PM.

Pickups concentrate in a few neighbourhoods

Indira Nagar, Banashankari, Ramamurthy Nagar and Kammanahalli topped the list, each with roughly 840–880 pickups in the month, followed by BTM Layout, Cox Town, Sarjapur Road and Kengeri. These are dense residential and commercial areas, several of them known for traffic congestion, which is consistent with people choosing a ride over driving. Operationally this is where driver positioning during peak hours would matter most.

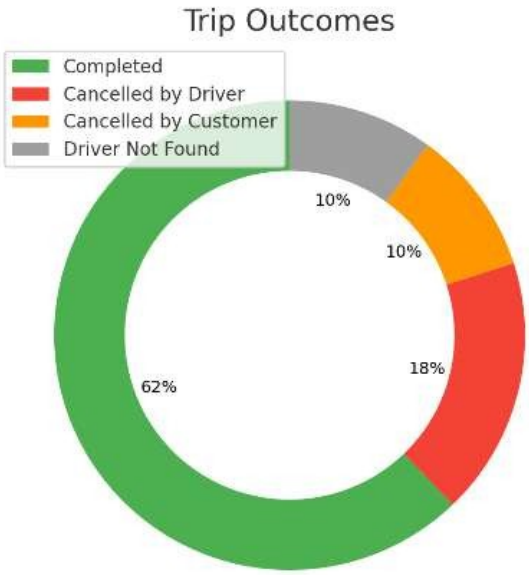


Top eight pickup locations by number of requests.

38% of requests never became a ride

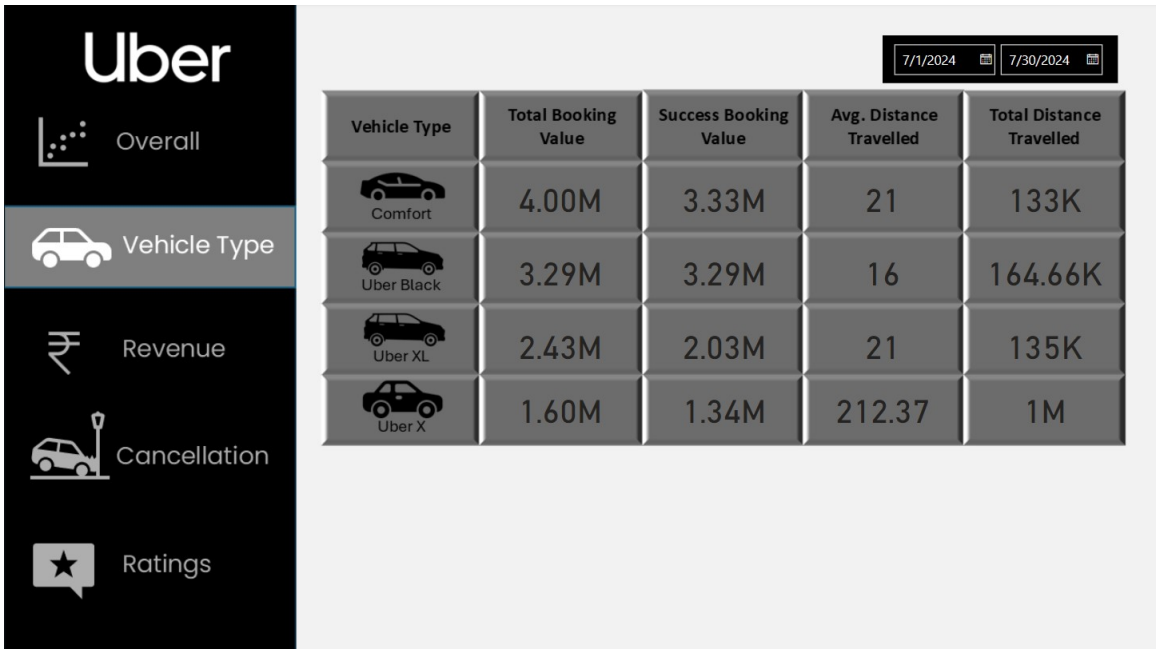
62% of bookings completed. Of the rest, 18% of all requests were cancelled by the driver, 10% by the customer, and 10% ended as Driver Not Found. That is a large reliability gap. A closer look showed Driver Not Found

clustering in peak hours in outlying areas, which points to supply being stretched rather than riders changing their minds.



Booking outcomes as a share of all requests.

Premium tiers carry the revenue



Vehicle Type	Total Booking Value	Success Booking Value	Avg. Distance Travelled	Total Distance Travelled
Comfort	4.00M	3.33M	21	133K
Uber Black	3.29M	3.29M	16	164.66K
Uber XL	2.43M	2.03M	21	135K
Uber X	1.60M	1.34M	212.37	1M

Booking value, completed-trip value, average and total distance by service tier.

Each tier had about a quarter of requests, but revenue split very differently. Comfort generated the most, over ₹3.3 million, on an average fare above ₹500. Uber Black followed at about ₹2.76 million. Uber X, the budget tier, averaged around ₹212 a trip and brought in about ₹1.34 million. Payments were mostly cash (about half of completed trips) and UPI (about 40%), with cards a small minority.

Technical notes

- **Pre-aggregating in SQL** rather than importing 40,000 raw rows and doing everything in DAX kept the report responsive and made the logic easy to audit.
- **Calendar table in Power BI** for weekday/weekend slicing.
- **DAX measures** for completion rate and cancellation share, so the percentages respond correctly to date, location and tier filters.
- **Version discipline.** The cleaned dataset used in Power BI is the same one produced by the SQL step; I kept one copy to avoid the two drifting apart.

What I took from it

On the business side: demand is predictable enough to plan around, the failure rate is the real problem, and a small set of neighbourhoods and hours is where driver supply decisions would pay off. Premium tiers are worth protecting because they fund the volume tiers.

On the technical side: the "No" placeholder issue and the corrupted location names were a good reminder that the first job is always to look at the data before modelling it. Fixing them in SQL, where the rule is written down, beat fixing them by hand in Excel.

Next steps if I extend this: more months to see seasonality, external factors like weather or holidays, and a simple model predicting cancellation likelihood per request.

The dataset is a learning dataset modelled on a public ride-hailing case study for Bangalore. It is not Uber's data, and the findings describe the dataset, not the company.